

EcoPackAI

Sustainability Report

AI-Powered Eco-Friendly Packaging Intelligence

Generated on 1 April 2026

Data Source: EcoPackAI PostgreSQL Database

Classification: Internal Report · Milestone 4 — Module 7

512

Total Materials

87.3%

Avg Biodegradability

0.852 kg/kg

Avg CO₂ Emission

\$2.01

Avg Cost / kg

46.7%

Avg Recyclability

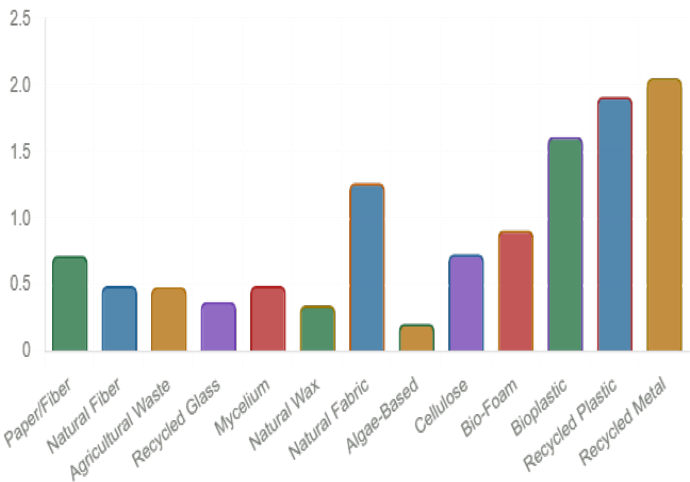
Top 20 Eco-Friendly Materials by Suitability Score

#	Material Name	Type	Source	Biodeg%	C O ,	Cost	Recycle%	Score
1	Molded Pulp (Anti-Static)	Paper/Fiber	Recycled	97%	0.40	\$0.84	78%	0.8
2	Hemp Fiber Board (Flexible)	Natural Fiber	Renewable	97%	0.63	\$2.38	71%	0.8
3	Jute Sacking (Coated)	Natural Fiber	Renewable	99%	0.40	\$0.89	63%	0.8
4	Wood Wool (Standard)	Natural Fiber	Renewable	99%	0.49	\$1.49	68%	0.8
5	Jute Sacking (Export-Grade)	Natural Fiber	Renewable	97%	0.56	\$0.49	61%	0.8
6	Hemp Fiber Board (Premium)	Natural Fiber	Renewable	97%	0.57	\$1.82	58%	0.8
7	Jute Sacking (Ultra-Thin)	Natural Fiber	Renewable	95%	0.37	\$0.57	72%	0.8
8	Jute Sacking (Eco-Plus)	Natural Fiber	Renewable	98%	0.39	\$0.68	69%	0.7
9	Bamboo Composite (Moisture-Proof)	Natural Fiber	Renewable	94%	0.41	\$1.57	54%	0.7
10	Kraft Paper (Food-Grade)	Paper/Fiber	Renewable	98%	0.62	\$0.76	78%	0.7
11	Jute Sacking (Flexible)	Natural Fiber	Renewable	97%	0.37	\$0.73	70%	0.7
12	Corrugated Cardboard (Reinforced)	Paper/Fiber	Renewable	97%	1.16	\$0.78	87%	0.7
13	Coconut Fiber Board (Anti-Static)	Natural Fiber	Renewable	97%	0.45	\$1.22	30%	0.7
14	Jute Sacking (Food-Grade)	Natural Fiber	Renewable	94%	0.51	\$0.81	76%	0.7
15	Corrugated Cardboard (Heavy-Duty)	Paper/Fiber	Renewable	87%	0.58	\$1.05	82%	0.7
16	Wood Wool (Premium)	Natural Fiber	Renewable	98%	0.73	\$1.33	71%	0.7
17	Molded Pulp (Heavy-Duty)	Paper/Fiber	Recycled	93%	0.76	\$0.89	72%	0.7
18	Jute Sacking (Premium)	Natural Fiber	Renewable	93%	0.37	\$0.62	72%	0.7
19	Bamboo Composite (Heavy-Duty)	Natural Fiber	Renewable	92%	0.26	\$2.24	55%	0.7
20	Bamboo Composite (Lightweight)	Natural Fiber	Renewable	89%	0.31	\$2.54	53%	0.7

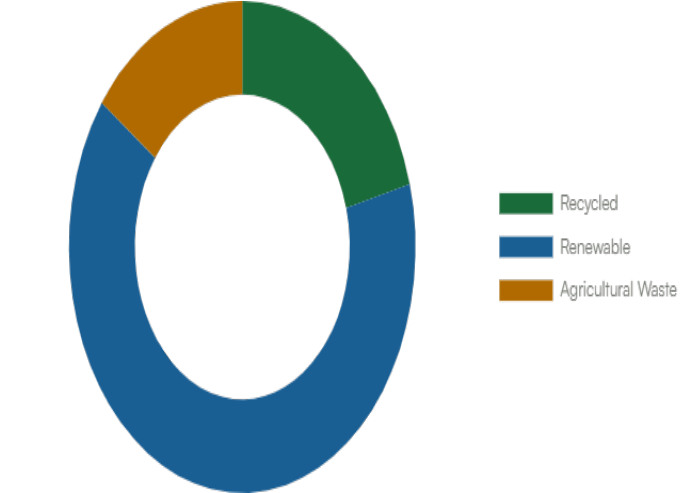
Sustainability Analysis by Material Type

Material Type	Count	A v g C O ,	Avg Biodeg%	Avg Cost	Avg Recycle%
Paper/Fiber	106	0.722	88.3%	\$0.78	74.9%
Natural Fiber	94	0.495	91.5%	\$1.65	54.3%
Agricultural Waste	77	0.485	92.8%	\$1.23	25.7%
Recycled Glass	15	0.373	69.0%	\$0.49	92.7%
Mycelium	15	0.495	94.1%	\$4.58	2.4%
Natural Wax	10	0.348	93.7%	\$5.35	4.2%
Natural Fabric	12	1.267	93.2%	\$3.13	52.3%
Algae-Based	27	0.210	95.6%	\$5.35	3.4%
Cellulose	12	0.732	89.8%	\$2.71	14.1%
Bio-Foam	27	0.913	86.5%	\$2.53	11.0%
Bioplastic	68	1.616	84.0%	\$2.79	29.3%
Recycled Plastic	39	1.917	69.0%	\$1.44	84.9%
Recycled Metal	10	2.056	69.0%	\$3.81	91.5%

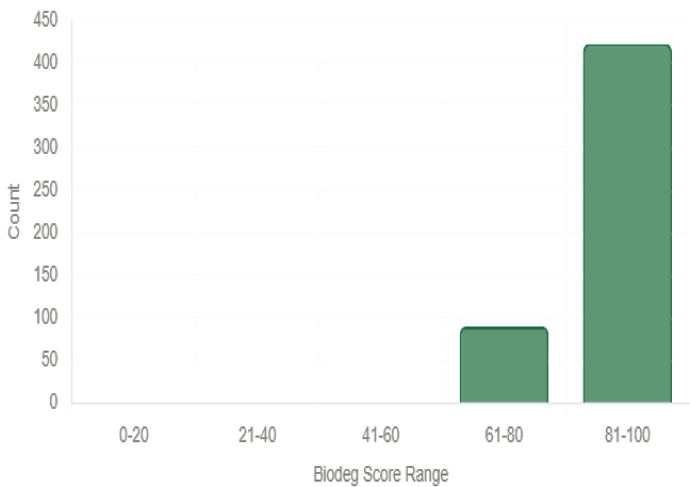
CO₂ by Type



Source Distribution



Biodegradability



Recyclability